PAPER 3 — HANDOFF v13.0

Written 28 August 2026. Supersedes v12.0 where they disagree. Keep v12.0 for contact-aware closing, the object library and the batch runner; v11.0 for the grid designer's transposition/palm/throat story; v10.0 for Block 2/3 and the diagonal-blob finding. None is repeated here in full.

0. WHERE THE PROJECT IS

Area	State
Cylinders	Done. Widths 18, 26, 40, 60 validated end to end (0 dead, indentation 1.3–1.5 mm). Remaining widths are repetition.
Cuboids	Path built. Scaled box primitive of any size; shape-aware band; edge-seeking grid. One 30x120x140 calibrated and swept. NOT yet validated with a firm grasp — see 5.1.
Spheres	Untouched. The band formula exists; nothing has been run.
Rolled / tilted data	Blocked on Berith's CNN. Collect 0 and 90 deg only.
Batch collection	Built and tested; not yet used at scale.

Verify file freshness before editing. Grep for a signature:

File	Signature
main_gui.py	contact_band_mm / _cal_key / OBJ_BOX
sim/collect_from_config.py	SCALED CUBOID / cal_key
viz/stitching.py	grip_mm / _load_dead_grasps
viz/heatmaps.py	whole {d:.0f} mm face
examples/probe_finger_throat.py	_keys / CAL_PATHS
Data/build_object_library.py	band_width_mm

1. SHAPES: ONE RULE, THREE BANDS

contact_band_mm(shape, width) in main_gui.py is the only genuinely shape-specific piece of design_grid, and getting it wrong is silent: the grid still runs, the maps still stitch, and the outer columns simply grasp less than intended.

Shape	Across band	Along band	Why
cylinder	sqrt(d(W-d))	none	curves one way; contact is a strip running the pad's full height
cuboid / cube	**W/2**	none	flat on flat — NO compliance limit; contact is the whole face
sphere	sqrt(d(W-d))	**same**	curves both ways, so the contact is a patch and ALONG is limited too

The cuboid band is much larger: at 18 mm the pad may travel **17.0 mm instead of 14.1**; at 40 mm, **28.0 instead of 17.5**. Reusing the cylinder formula under-swept every cuboid.

The band uses the ACROSS extent, not the grasped width. On a cuboid these differ completely, and passing the grasped width would give every one of Berith's cuboids the same band.

1.1 The three axes are fixed by the scene

Axis	Meaning	Used for
world X	what the JAW closes on	calibration + throat key
world Y	ACROSS the pad	contact band, object outline, expected-band panel
world Z	ALONG the pad	pad overhang, throat depth, stand-on-table

The library used to take the grasped width as min(dims[0], dims[1]), harmless for a cylinder and wrong for a cuboid. **Cuboid_M is 40 x 18 x 100**: the library called it 18 mm, the gripper closed to an 18 mm gap on a 40 mm object and drove the pads 22 mm INTO it (indentation 2.04 mm instead of ~1.3). The same bug on the other axis put the _H variants 41 mm above the table.

> ***A useful consequence:*** *all five of Berith's cuboids are 40 mm in the closing direction — his 8/18/40 ladder is the ACROSS dimension. So one 40 mm calibration unlocks every one of them, and what varies is how much face the pad sees: 8 mm (well inside the 22 mm pad), 18, and 40 (wider). That spans exactly the regime where Paper 1's edge-mirroring rule starts to apply.*

2. THE SCALED CUBOID

A box PRIMITIVE, not a mesh — no USD from anyone, and any Paper 1 geometry can be built by typing three numbers. Second entry in the object dropdown: **(scaled cuboid — type W x D x H)**.

Field	Axis	What it decides
W (grip)	X	the jaw gap -> which calibration and throat entry
D (across)	Y	the contact band -> the whole sweep
H (along)	Z	overhang + throat limits, and the table height

Emitted as GRASP_OBJECT_SHAPE=cuboid and GRASP_OBJECT_BOX_MM=W,D,H; the collector defines a UsdGeom.Cube, scales it, applies CollisionAPI, puts its base on Object_02's origin and deactivates the unit cylinder.

Each shape owns its own size fields. box_w/box_d/box_h for the cuboid, obj_diam/obj_len for the cylinder; only the SELECTED shape feeds CYL_D/CYL_L. Before this, typing a cuboid width silently changed the cylinder's diameter and there was no way to tell which was in force. Typing is debounced 400 ms — applying on every keystroke redrew "140" as 1 mm, then 14, then 140.

2.1 Getting an EDGE into the map

A sweep that sits entirely on a flat face produces a featureless plateau and teaches a completion model nothing about where the object ends. Three things now make an edge reachable:

1. **The canvas caps the ACROSS count.** The along count always was capped this way; across never was, because a cylinder's band cannot get close to 96 mm. A 120 mm face allows 154 mm of sweep, so the design was REFUSED at the very end instead of trimmed.
2. **"let the pad hang past the ENDS"** (checkbox, default OFF). The two directions were asymmetric: across, the pad may hang off with across_margin_mm still touching — that is what puts an edge in the map; along, it had to stay wholly on the body, so the END could never be seen AT ANY LENGTH. Geometry never forbade it: the throat floor sits below that ceiling for every length. On 30x120x140 this takes the sweep from 91 points reaching +65.1 mm to 117 points reaching +88.1 mm against a body end at +70.
3. **"auto y anchor"** (checkbox, default OFF). Centring leaves a wide face showing only plateau; a hand-typed y overshoots (y=45 on a 120 mm face put the outer column's pad entirely past the edge, touching nothing). Derived instead: y = across_max - n_across * step, which lands the far column exactly on the across limit. Verified across four face widths — every one leaves exactly 3.00 mm of pad in contact.

With both boxes ticked and y/z left blank, one 96 mm canvas straddles the SIDE edge and the TOP end on any size. Both default off, so every cylinder design ever made is unchanged.

> *Two earlier claims of mine that were WRONG and are corrected here: "you cannot have both edges" and "make the body 80 mm". Height never mattered; the blocker was the along-direction rule, not the geometry.*

3. CALIBRATION IS KEYED BY SHAPE

Key format: bare width for a cylinder, "width|shape" otherwise. So 26.0 and 30.0|cuboid. Cylinders keep the bare key, so every entry measured before shapes existed still reads. Used identically by the GUI, the collector, the throat probe and the object library.

There is NO cross-shape fallback, deliberately. A fixed-angle close_rad DOES transfer between shapes — same jaw gap — but a contact-aware one does NOT, because a flat face trips the target far earlier than a round one. The file does not record which method produced an entry, so borrowing is unsafe. Both the GUI and the collector refuse and say so: *"a CYLINDER entry for this width exists and is deliberately NOT used"*. A warning buried in a 400-line log is not a decision.

3.1 Use DEFORMATION, not tactile counts

Contact-aware calibration: signal = deformation (mm), target = 1.40, for every shape and every width.

Tactile counts cannot be the uniform basis across shapes. At the SAME physical indentation a flat 30 mm face gives far more contact area than a 30 mm cylinder, so the same count target is a much lighter squeeze on the cuboid. Observed: a 12000-count target on the 30x120x140 cuboid tripped early, stored a small close_rad, and every grasp since was too weak to register properly. Raising it to 140000 does not help — it simply never trips and closes to the ceiling.

Indentation is the physical quantity that transfers: 1.4 mm is 1.4 mm on a flat face or a round one, and it is the value already validated across nine cylinder widths.

Tactile remains available and correct for CYLINDERS, where the 8600–15400 band was established. It is the CROSS-SHAPE comparison it cannot support.

4. THE WORKFLOW FOR ANY NEW OBJECT

#	Step	Where	When
1	Pick or build the object	Collection tab	always
2	Calibrate — mode **both (compare)**, signal **deformation**, target **1.40**	Calibrate tab	once per shape+width
3	Throat probe	button 1 -> terminal	after any NEW shape+width
4	Rebuild object library	button 2	only for Berith's MESHES; a scaled box is not in the library
5	Design grid	Collection tab	always

Step 3 is not optional and not automatic. design_grid reads finger_throat.json by shape+width; without an entry it refuses. The probe reads close_rad per entry from all three calibration files, so it can only measure what is already calibrated — that is why the order matters. It writes finger_throat.json under the same keys.

> ***Calibration Z offset*** *is typed by hand; it only has to put the pad on the body clear of everything. In CALIBRATE mode the EE uses the PROVISIONAL TOOL_OFFSET_Z = 0.150 (the real one is what it is measuring), so the pad lands ~6.5 mm below the typed target. That is expected, not a fault.*

4.1 Which calibration file collection uses

A dropdown on the Collection tab picks main / fixed / contact, and it drives BOTH the grid designer and the run command — they must agree, since the offsets differ by up to 0.73 mm between methods. **Default is contact.** Mixing methods inside one dataset is what ruins a week.

5. OPEN ITEMS, IN PRIORITY ORDER

5.1 Re-calibrate the 30 mm cuboid on deformation

The immediate next action. The existing 30.0|cuboid entry was measured with a tactile target and grips too weakly to collect with. Redo it with deformation / 1.40, then throat probe, then sweep. Expect indentation ~1.3–1.5 mm at the hold and a tactile peak in the thousands, matching the cylinders.

5.2 No ACROSS-SIDE limit exists

design_grid checks the pad against the object's ENDS (overhang) and against the gripper (throat, palm), but nothing checks that an across anchor keeps the pad on the object at all. "auto y anchor" derives a safe value, but a hand-typed y is unchecked: on a 120 mm face, y beyond about 55 runs the pad off entirely. This is the one place a silently-bad grid can still be produced.

5.3 A refused design does not block Save Config

Design Grid can report REFUSED and the run command is still offered, so a rejected grid can be collected. Observed on the 30x120x140 cuboid before the canvas cap existed.

5.4 Old run folders draw the wrong object

The config used to write `"shape": "cylinder"` hardcoded and recorded only the grasped width, so heatmaps, stitching and blob were all told a cuboid was a Ø30 cylinder: the outline came out 30 mm wide around a 120 mm object, and the EXPECTED panel predicted a 16.3 mm chord band the pad — anchored 32 mm off centre — never touched, hence an empty panel. Fixed at source: the config now records the real shape and an **across_mm**, _load_scene returns both, and the expected band uses the whole face for a flat shape. **Runs collected BEFORE this still replot wrong**, because their config carries the old block.

5.5 Sphere path is untested

contact_band_mm handles spheres and the along-band limit works ("bound by contact band" appears), but no sphere has been calibrated or grasped. Berith's "spheres" are 40x40x90 and 65x65x90 — capsules, not spheres — so check one visually before assuming Paper 1's spherical rule applies.

5.6 Cuboid yaw is undefined

The scene applies no yaw, so the pads meet the +X/-X faces of an axis-aligned box. That is the correct default for a FACE grasp, but nothing records it, and an EDGE or corner grasp is a completely different tactile signature that Paper 1's mirroring rule does not cover.

5.7 Carried forward from v12.0

- **Diagonal blob artifact** — Berith confirmed it is real and is working on it. The four live captures at 0/20/45/90 deg are the clearest evidence: 0 gives a clean vertical ridge and 90 a clean horizontal one, while 20 and 45 collapse into broad blobs — 45 deg has the HIGHEST sum (12195) with the LOWEST peak (999). Collect 0 and 90 deg only.
- **100 mm vs 140 mm bodies** — the six earliest sweeps are 140 mm scaled cylinders; Berith's objects are 100 mm. Not the same object; do not pool them without saying so.
- **Plots draw shapes but not spheres** — top-down, front, 3D and the calibration panel are all shape-aware for cylinder and cuboid.
- **validation.py** still assumes an axis-aligned pad footprint; correct on flat runs only.
- **GSR saturated** — no dynamic range; do not quote it.

6. CONSTANTS AND FORMULAS

Name	Value	Provenance
PALM_DROP_MM	86.69	measured: 10.79 + 75.9
THROAT_MARGIN_MM	1.0	bracketed: dead -0.06, D26 lowest +1.88, calibration pose +3.13
DEAD_GRASP_FLOOR	0.0002 m	0.2 mm; rest 0.000, firm 1.3–1.5
CONTACT_TARGET default	0.0014 m	1.4 mm
across_margin_mm	3.0 mm	pad still touching at the outermost column
INDENT_MM	2.4 mm	compliance model; 2.09 measured on D60
canvas_mm	96.0	4 halvings: 96/48/24/12/6
PAD_W, PAD_H	22.0, 37.0 mm	4 cols x 5.5, 7 rows x 5.29
OBJ_BASE_Z	982.2 mm	obj_z = 982.2 + L/2
grid step default	6.0 mm	6 - 5.5 = 0.5 mm of sub-taxel shift per step

design_grid's six limits, in the order they are applied: (1) object big enough to indent; (2) ACROSS — pad_hw + band - margin, capped by the canvas; (3) ALONG floor — throat, so the object's top end cannot jam in the fingers; (4) ALONG ceiling — pad stays on the body, or hangs past it if end_overhang; (5) canvas; (6) PER-POINT — the measured gripper cloud at that point's own rolled tool pose, 1 mm clearance. If a grid fails it re-centres the height, then trims a step, then REFUSES.

7. FILE INVENTORY

File	Role / recent change
main_gui.py	contact_band_mm; scaled cuboid + its own size fields; shape-keyed calibration; auto y anchor; end overhang; canvas cap on across; shape-aware previews (top-down, front, 3D, calibrate); config records shape + across_mm; debounced typing
sim/collect_from_config.py	scaled cuboid primitive; shape-keyed cal_key with no cross-shape fallback; contact-aware close; live publisher; external USD loading; placement by measurement; ground-plane fallback; ascent contact zone; log_mesh
viz/stitching.py	_load_scene returns shape, grip_mm and d (= across); drops dead grasps
viz/heatmaps.py	expected band uses the whole face for a flat shape
examples/probe_finger_throat.py	iterates calibration ENTRIES (shape+width), merges all three calibration files
Data/build_object_library.py	axis mapping X/Y/Z; band_width_mm; shape-keyed lookups
examples/run_batch.py	unattended queue, resume, all-dead = FAILED
examples/measure_objects.py	bbox of every USD

8. HOW TO WORK ON THIS

- One verifiable step at a time. Measure, don't guess. Distances in mm.
- Complete files, never diffs. Commands on one line.
- **Kourosh's visual read of the simulation overrides diagnostics when they conflict.** It has been correct every time — the wobbling object, the pad off the end of a short body, the pads inside the cuboid, and the cylinder still showing in the previews.
- Bad data must refuse LOUDLY rather than degrade silently.

The pattern behind almost every fault in this project: a wrong number that looked normal. A hardcoded "cylinder" in a config; a grasped width taken as the minimum dimension; a position inherited from a scene authored for a different length; a statistic measuring rigid motion instead of squash; a stale label; a calibration key that two shapes shared. None of these threw an error. Every check added since prints WHAT IT MEASURED, not just its verdict — keep that up.